

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Find the solution of the initial value problem in explicit form and determine the interval in which the solution is defined.

$$y' = (1 - 2x)y^2, \quad y(0) = -1/6.$$

Solution: $\frac{dy}{dx} = (1 - 2x)y^2$ $y(0) = -1/6 \Rightarrow -\frac{1}{6} = -\frac{1}{C}$

$$\frac{1}{y^2} dy = (1 - 2x) dx \quad \Rightarrow C = 6$$

$$\int \frac{1}{y^2} dy = \int (1 - 2x) dx$$

$$-\frac{1}{y} = x - x^2 + C$$

$$1/y = x^2 - x - C$$

$$\Rightarrow y = \frac{1}{x^2 - x - C}$$

Thus $y = \frac{1}{x^2 - x - 6}$

$$x^2 - x - 6 \neq 0$$

$$(x - 3)(x + 2) \neq 0$$

$$x \neq 3, x \neq -2.$$

Three intervals: $(-\infty, -2)$, $(-2, 3)$, $(3, \infty)$

pick up $(-2, 3)$ since 0 falls in that interval.

Problem 2. (4 points) Use the substitution $v = \frac{y}{x}$ to solve the differential equation

$$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2}$$

Solution: $\frac{dy}{dx} = 1 + \left(\frac{y}{x}\right) + \left(\frac{y}{x}\right)^2$

let $v = \frac{y}{x}$, then $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$

Thus

$$v + x \frac{dv}{dx} = 1 + v + v^2$$

$$x \frac{dv}{dx} = 1 + v^2$$

$$\frac{1}{1 + v^2} dv = \frac{1}{x} dx$$

$$\Rightarrow \int \frac{1}{1 + v^2} dv = \int \frac{1}{x} dx$$

$$\arctan v = \ln|x| + C.$$

$$v = \tan(\ln|x| + C)$$

$$x \neq 0.$$

$$\frac{y}{x} = \tan(\ln|x| + C)$$

$$\Rightarrow y = x \tan(\ln|x| + C)$$